Geonwoo Bang

ETH Zürich, Zürich, Switzerland
g7199@g.skku.edu ◇ Portfolio

EDUCATION

ETH Zürich **Sep 2026 – Present**
MSc in Data Science

Sungkyunkwan University **Mar 2020 – Feb 2026**
Triple Degree Program, Summa Cumme Laude
B.S. in Computer Science and Engineering 4.44/4.5
B.Econ. in Statistics 4.5/4.5
B.S. in Computer Education 4.21/4.5

PUBLICATIONS

Bang, G., et al. "Propensity-Weighted Reflection for Recovering Poisoned Recommenders under Partial Identification" *The ACM International Conference on Web Search and Data Mining (WSDM) 2027 accepted.*

Seo, J., **Bang, G.**, et al. "Whose Fairness? Auditing Exposure Fairness When AI Agents Read Rankings" *The ACM International Conference on Web Search and Data Mining (WSDM) 2027 accepted.*

Bang, G., et al. "EVOL: Simulator-Guided Evolutionary Expert Synthesis for Deployment-Free Learning Path Recommendation" *The Conference on Information and Knowledge Management (CIKM) 2026 accepted.*

Seo, J., **Bang, G.**, et al. "ReForest: Interpretability Meets Reasoning in Isolation Forest" *The Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD) 2026.*

Bang, G., et al. "Adaptive PII Detection via Evolutionary Orchestration and Multi-Agent Debate" *The Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD) 2026.*

Bang, G., et al. "Transforming User-Defined Criteria into Explainable Indicators with an Integrated LLM-AHP System." *Arxiv Preprint, 2025.*

Bang, G., et al. (2025). "SNAP: Enhancing Long-Form Narrative Agents with Cell-Based Segmentation and Plan-Driven Dialogue Strategies." *Conference of Korean Artificial Intelligence Association 2025*

Bang, G., et al. (2024). "Methodology for Implementing a Game-based Digital Therapy Using Generative AI to Reduce Impulsivity in Game-addicted Adolescents." *Journal of the Korea Institute of Information and Communication Engineering (JKIICE).*

RESEARCH EXPERIENCE

Motion AI Lab, Sungkyunkwan University **Mar 2025 - Present**
Research Assistant
Implemented real-time responsive motion matching system using motion capture data
Developed keyframe extraction and motion compression using diffusion and reinforcement learning

LAMDA Lab, Sungkyunkwan University **Jan 2024 - Jan 2025**
Research Assistant
Core team member in securing $5M national R&D grant on AI-based digital therapeutics
Developed generative AI NPC framework and co-authored 3 publications and a patent

PROFESSIONAL EXPERIENCE

Communications & IT Specialist **Feb 2022 - Oct 2023**
PO2 @ Republic of Korea Navy 2nd Fleet, Incheon, South Korea
Operated military-grade satellite, microwave, and radio communication systems
Deployed twice for North Korean drone threat response, ensuring flawless communication

PROJECTS

DrawMind: Gamified Art Therapy Platform for Mental Health Assessment **2025**
Team Leader, Gold Prize @ SKKU AI Hackathon 2025
Developed game-based drawing system integrating clinical psychology literature via Dynamic Context Injection
Enhanced diagnostic accessibility through Image-to-Image Generation providing immersive visual feedback

UniScore: Universal Explainable Text Scoring via AHP-JSD **2024–2025**
First Author, SWlab @ SKKU
Developed extreme-value aggregation framework achieving 107% correlation improvement over GPT-5 baseline
Reduced evaluation time by 86% while maintaining high interpretability through mathematically grounded weighting

LiterALL: AI-Powered Literacy Support Platform **2024**
AI Developer, Grand Prize @ SKKU AI Hackathon
Designed an AI-based framework for real-time text transformation and readability adaptation
Built personalized reading difficulty calibration and integrated gamification for engagement enhancement

Social Isolation Risk Mapping **2023**
Lead Data Analyst, Excellence Prize at Data Safety Zone Competition
Developed AHP-JSD, a novel scoring framework combining Analytic Hierarchy Process with Jensen-Shannon Distance
Applied to telecom-financial data to identify youth isolation clusters with interpretable social risk indices

AWARDS & HONORS

Award & Competition	Organizer	Year
Gold Prize – SKKU AI Hackathon	SKKU	2025
Grand Prize – SKKU AI Hackathon	SKKU	2024
Special Prize – K-PaaS Service Development Contest	MSIT, Korea	2024
Third Prize – MAFA Public Data Utilization Startup Contest	MAFA, Korea	2024
Excellence Prize – Data Safety Zone Competition	MSIT, Korea	2023

Scholarship	Institution	Year
Academic Excellence Scholarship (Top $\leq 4\%$)	SKKU	2024-25
Challenge Semester TA Scholarship	SKKU	2024
University Innovation Scholarship	SKKU	2024

TEACHING EXPERIENCE

Course	Institution	Role	Students	Term
Information and Communication	Sunrin HS	Instructor	20	Spring 2025
AI Basics & Uses	SKKU	TA	618	Fall 2024
Thesis Writing in Social Sciences using Python	SKKU	TA	524	Summer 2024
AI Basics & Uses	SKKU	TA	610	Spring 2024
AI Basics & Uses	SKKU	TA	589	Fall 2023